

LINUX FUNDAMENTAL

Introduction

Linux is a family of open-source Unix-like operating system based on the Linux kernel. This module covers some of the history behind Linux, where an OS kernel was first released on September 17, 1991, by Linus Torvald. We shall also learn and run our very first commands in an interactive Linux machine. The module provides some essential commands used to interact with file systems. As penetration testers we shall have an understanding of how users and groups work on Linux and how this means to us.

Level 1.

This level provides a bit background on Linux. Got familiar with where Linux is used as it powers things such as websites visited, point of sale system (pos) and critical infrastructure. Flavors of Linux were also discussed, and a good example of the same is Debian and Ubuntu.

Answer the questions below

Research: What year was the first release of a Linux operating system?

1991

✓ Correct Answer

Level 2.

Deployed and interacted with a Linux machine in my browser.

```
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.11.0-1017-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

System information as of Mon Jun 10 03:37:08 UTC 2024

System load:  0.12           Processes:            111
Usage of /:   18.8% of 9.63GB Users logged in:          0
Memory usage: 39%           IPv4 address for ens5: 10.10.135.
252
Swap usage:   0%

0 updates can be applied immediately.

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

tryhackme@linux1:~$
```

Answer the questions below

I've deployed my first Linux machine!

No answer needed

✓ Correct Answer

Level 3.

Used a Terminal to run my first commands, i.e “**echo**” used to output any text provided. And “**whoami**” to find what user I am logged in as.

```
tryhackme@linux1:~$ echo "tryhackme"
tryhackme
tryhackme@linux1:~$ whoami
tryhackme
tryhackme@linux1:~$ █
```

If we wanted to output the text “**TryHackMe**”, what would our command be?

echo TryHackMe

✓ Correct Answer

💡 Hint

What is the username of who you're logged in as on your deployed Linux machine?

tryhackme

✓ Correct Answer

💡 Hint

Level 4.

On the Linux machine that you deploy, how many folders are there?

4

✓ Correct Answer

Which directory contains a file?

folder4

✓ Correct Answer

💡 Hint

What is the contents of this file?

Hello World

✓ Correct Answer

Use the cd command to navigate to this file and find out the new current working directory. What is the path?

/home/tryhackme/folder4

✓ Correct Answer

Interacted with the file system by learning and running commands such as **ls** – used for **listing**, **cd** – for **changing directory**, **cat** – **concatenate**, **pwd** – **print work directory**.

Level 5.

Covered more commands, “**find**” and “**grep**”. “find” command is used to automate very simply or complex finding of what to be extracted. Can also be used with a wildcard (*) to anything with a particular extension. “grep” allows us to search the content of files for specific values being looked for.

```
tryhackme@linux1:~$ grep "THM" access.log
13.127.130.212 - - [04/May/2021:08:35:26 +0000] "GET THM{ACCESS} lang=en HTTP/1.1" 404 360 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/77.0.3865.120 Safari/537.36"
```

Use grep on "access.log" to find the flag that has a prefix of "THM". What is the flag?

THM{ACCESS}

✓ Correct Answer

💡 Hint

And I still haven't found what I'm looking for!

No answer needed

✓ Correct Answer

Level 6.

This level covered some shell operators. The following were cover in this module.

& - allows us to run commands in the background of our terminal

&& - allows combination of multiple commands together in one line of the terminal. Eg. Cmd1 && cmd2, where cmd1 must be successful.

> - known as a output redirector. means is that we take the output from a command we run and send that output to somewhere else eg. Echo hey > welcome

>> - same as (>), but instead, it appends the output rather replacing i.e nothing is over written.

If we wanted to run a command in the background, what operator would we want to use?

&

✓ Correct Answer

If I wanted to replace the contents of a file named "passwords" with the word "password123", what would my command be?

echo password123 > password

✓ Correct Answer

💡 Hint

Now if I wanted to add "tryhackme" to this file named "passwords" but also keep "passwords123", what would my command be

echo tryhackme >> password:

✓ Correct Answer

💡 Hint

Now use the deployed Linux machine to put these into practice

No answer needed

✓ Correct Answer

Conclusion

This lab explored the fundamental of linux in depth. We were able to understand why linux is so common, interacted with a linux machine and ran some of the most fundamental commands. This lab also provided an introduction to navigating around the filesystem and how we can use commands like find and grep to make finding data even more efficient. Lastly covered important shell operators.



Congratulations!

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